

RUGWED PATHARKAR

AI Engineer · Backend & Agentic-AI Platforms

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Professional Summary

AI Engineer, 2.5+ years shipping production LLM-powered, multi-tenant agentic microservices for hospitality SaaS. Owner of a 31-service Python/Go/FastAPI/gRPC platform on GKE with multi-region deployments, 12+ third-party integrations (10 PMS, door-lock, GRMS), a Vapi-based voice-agent runtime, and a 10-specialist multi-agent supervisor across Claude 3.7 Sonnet, Gemini 2.5, GPT-4.1, and Groq Llama with MCP tool-calling, evals, and hybrid dense+sparse RAG. Cut p95 API latency 96% (5s to 200ms); Star Performer of the Quarter and consistently top-ranked engineer at Upswing (9.0+ average across every performance review).

Skills

Languages: Python, Go, JavaScript, TypeScript, Java, Protocol Buffers, SQL, Shell Scripting

AI & Agentic AI: LangChain, LangGraph, Model Context Protocol (MCP), MCP Server Development, Multi-agent Systems, Agent Orchestration, Prompt Engineering, LLMops, RAG, Hybrid Retrieval (Dense + BM25/Sparse), Vector Search, Semantic Search, Embeddings, FastEmbed, Claude 3.7 Sonnet, GPT-4.1, Gemini 2.5, Groq Llama 3.3, Google Vertex AI, Google ADK, Vapi (Voice AI), Deepgram, ElevenLabs, Prophet, Model Routing

Evals & Observability: LangSmith, ragas, promptfoo, structured Pass/Fail rubric analysis, JSON-schema intent/sentiment/outcome extraction, golden-payload drift tests, per-tool SLO deadlines, Prometheus, Grafana, custom watchdog exporters

Backend: FastAPI, gRPC, Django, Spring Boot, Node.js, Celery, WebSockets, Socket.IO, FastMCP, Pydantic, asyncio

Frontend: React, Next.js, Vite, Tailwind CSS, Redux, Three.js, Angular, Nx

Databases & Search: MongoDB, PostgreSQL, MySQL, Firebase Firestore, Redis, Google BigQuery, ChromaDB, Qdrant (Vector DB)

Distributed Systems: Event-driven Architecture, Idempotency, Eventual Consistency, Multi-tenancy Isolation, Caching, Connection Pooling, Multi-region Deployments, Observability, On-call and Incident Response

Cloud & DevOps: GCP (GKE, BigQuery, Pub/Sub, Cloud Storage, Vertex AI), Docker, Kubernetes, Helm, Helmfile, Terraform, Pants, Apache Airflow, GitHub Actions, GitLab CI/CD, Prometheus, Grafana, Trivy, CI/CD Pipelines

Messaging & Integration: RabbitMQ, Google Pub/Sub, MQTT, Stripe, SMTP, Resend, Zoho CRM SDK, Webhooks, REST API, SOAP API, XML

Testing & Practices: pytest, Integration Testing, Smoke Testing, CI Quality Gates, Code Review, TDD, Agile

Security: JWT, Firebase Authentication, OAuth 2.1, RBAC, Multi-tenancy Isolation, AES Encryption

Work Experience

Software Engineer · *Upswing Cognitive Hospitality Solutions*

May 2024 – Present

- Architected and own end-to-end a 31-service, multi-tenant Python/Go/FastAPI/gRPC backend on GKE with multi-region production deployments, including an API Gateway translating external REST to internal gRPC over Protocol Buffers.
- Built a Go voice-agent runtime on Vapi (Deepgram nova-3 STT with autoFallback, ElevenLabs TTS, smart-endpointing, structured analysisPlan for per-call intent/sentiment/outcome extraction) exposing 7 MCP tools with per-tool 15s context deadlines and tenant isolation via HTTP header injection — blocking LLM-argument-based cross-org impersonation.
- Built a production multi-agent supervisor coordinating 10 domain specialists (housekeeping, complaints, maintenance, F&B, inventory, guest requests, lost-and-found, insights, auto-assign, team) with streaming single-token classification for sub-1s first-token latency, MongoDB-checkpointed state, MCP tool-calling, dynamic routing across 4 LLM providers (OpenAI, Gemini, Claude, Groq), LangSmith observability, and dual-store session persistence (Mongo full-context + Firestore FE-typed payloads).
- Rewrote the guest-facing conversational orchestrator in Go with a 3-iteration singleflight-guarded specialist loop, a reconnecting RabbitMQ topic-exchange RPC transport (panic-isolated consumers, reply-envelope publishing), per-org prompt hot-reload via version-counter bumps, Redis fail-open dedupe on cross-service context sync, and a chain-of-thought leak filter — shaving 1.5–6s per response via verbatim-bypass and short-circuit synthesis paths.
- Evolved the RAG stack from ChromaDB to Qdrant with server-side hybrid search — RRF fusion over dense embeddings and BM25/FastEmbed sparse vectors via Prefetch + FusionQuery, lazy IDF Modifier installation, character-based semantic chunker (section > paragraph > sentence break-points), and multi-tenant collection isolation with multi-region write mirroring.
- Consolidated 10 vendor PMS integrations (Apaleo, Opera OWS, OHIP, Cloudbeds, Mews, RMS, HOP, Maxxton, StayNTouch, Clock) behind an abstract Reservation/Guest/ARI base-class hierarchy with a

shared exponential-backoff HTTP retry engine + per-provider webhook-health SLO tracking — new vendors drop to a bounded ~200-400 LOC transform-only change, cutting redundancy 60%.

- Built idempotent webhook processing (FastAPI BackgroundTasks + Mongo find_one_and_update upserts + Redis-locked reservation sync with atomic GETDEL Lua) and enterprise door-lock (ASSA ABLOY, Messerschmitt) + Guest Room Management System integrations via a Socket.IO namespaced real-time server, split cloud-gRPC + on-prem-HTTP topology, and Google Pub/Sub subscription filtering for room-lifecycle events.
- Cut p95 API latency 96% (5s to 200ms) via Redis caching, connection pooling, query optimization, and efficient serialization (~25% compute cost cut); migrated production Redis to a multi-shard Redis Cloud cluster and made the shared client cluster-safe (cross-slot handling), fixing distributed race conditions in inventory-hold and pricing under concurrent booking flows.
- Engineered event-driven architecture (RabbitMQ + Google Cloud Pub/Sub, at-least-once + idempotent) across a multi-protocol layer (REST, gRPC, WebSockets, SSE, Socket.IO); designed a MongoDB + BigQuery dual-write strategy powering automated Excel/PDF reports and 8 Airflow DAGs (document/media ingest, LLM crawler, hourly/daily analytics).
- Deployed self-hosted GitHub Actions runners with a matrix-fanout CI walking `src/` for changed Python + Go services (pytest + golden-payload drift tests, ~65% coverage, Trivy scanning, AES-128 at rest); provisioned Terraform + Helm/Helmfile + Pants infrastructure on Prometheus/Grafana (custom watchdog_exporter for multi-org voice-agent reachability) with production on-call and ~99.9% availability.
- Led and mentored a team of 3+ engineers while sustaining ~200 PRs/quarter with 100% peer-review coverage and 0 reverts, establishing reusable architecture patterns and code-review standards across multiple codebases.

IT Trainee Intern · Tech Entrepreneur

Mar 2023 – Sept 2023

- Engineered 10+ filtering categories for a job-matching portal (95% filter accuracy) with an advanced sorting system that cut candidate/job-filtering time 30%; ran regular code reviews enforcing best practices (20% code-quality improvement) and translated business requirements into actionable tasks.

Projects

- **Multi-Agent Housekeeping Supervisor** — Hand-rolled supervisor coordinating 10 domain specialists (housekeeping, complaints, maintenance, F&B, inventory, guest requests, lost-and-found, insights, auto-assign, team) with streaming single-token classification for sub-1s first-token latency; MCP tool-calling across 4 LLM providers (OpenAI, Gemini, Claude, Groq) with dynamic model routing, LangSmith observability, MongoDB-checkpointed conversational state, and dual-store session persistence.
- **Conversational Multi-Agent Orchestrator (Go)** — Latency-critical Go rewrite of the guest-facing chat orchestrator (~5,200 LOC, 92 .go files) — 3-iteration singleflight-guarded specialist loop, RabbitMQ topic-exchange RPC with reply-envelope publishing, per-org prompt hot-reload, Redis fail-open dedupe, chain-of-thought leak filter.
- **Enterprise Integration Platform** — Provider-agnostic backend for 10 Property Management Systems + enterprise door-locks (ASSA ABLOY, Messerschmitt) + Guest Room Management Systems; abstract Reservation/Guest/ARI base classes, shared HTTP retry engine, Fernet-encrypted credentials, idempotent webhooks; 60% less redundancy, 50% faster onboarding, 96% lower p95 latency.
- **AI Interview Platform (personal)** — Full-stack agentic interview system: LangGraph + Gemini + LiveKit real-time voice interviewer, two MCP servers (data + capability with Qdrant + Gemini-embeddings RAG), Python 3.12 backend on FastAPI + gRPC/gRPC-web, TypeScript pnpm+turbo frontend monorepo with unified candidate/recruiter experience, OpenTelemetry observability.
- **Custom MCP Server Suite** — 5 standalone MCP servers — Redis Sentinel dev-loop, multi-target RabbitMQ, Pants service-runner (13 tools + kubectrl port-forward automation), guest-booking + Stripe (27 tools), voice-agent provisioning — for Kubernetes-native local dev with async tool registries.
- **Portfolio Website · live at rugwedpatharkar.vercel.app** — Scientifically-accurate 3D solar-system portfolio: React 18 + Vite 5 + Three.js with @react-three/fiber, real NASA/JPL orbital data, cinematic scroll-driven tour with Lenis; Vercel deploy.

Education

MSc (Computer Applications) · Savitribai Phule Pune University · 81.95%

2021 – 2023

BSc (Computer Science) · Savitribai Phule Pune University · 72.57%

2017 – 2021